BLUESTOCK FINTECH

Mutual Fund Data Analytics Capstone Project — Final Report

Data Collection • ETL • Financial Performance Analytics • Power BI Dashboards

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Role	Data Analyst — Python | SQL | Power BI | Excel
Project Duration	June 20, 2026 – July 10, 2026
Submitted To	Yash Kale (Project Manager), Bluestock Fintech

Table of Contents

01 Project Overview & Objectives

This project is a comprehensive Mutual Fund Data Analytics solution developed as part of the Bluestock Data Analytics Capstone. It focuses on collecting, cleaning, storing, analyzing, and visualizing mutual fund data to generate meaningful business insights for investors and asset management stakeholders.

The project follows the complete data analytics workflow:

- Data Collection
- Data Cleaning & ETL
- Database Design
- Exploratory Data Analysis (EDA)
- Financial Performance Analytics
- Advanced Analytics
- Interactive Dashboard
- Reporting

Objectives

- Build an automated ETL pipeline for mutual fund datasets.
- Store cleaned data in a SQLite database.
- Analyze fund performance using financial risk and return metrics.
- Visualize industry, performance, and investor trends using Power BI.
- Recommend funds to investors based on risk and performance profile.
- Generate actionable insights for investors and stakeholders.

Project Timeline

Start Date	June 20, 2026
Submission Date	July 10, 2026
Prepared By	Shaik Mohammad Vali
Submitted To	Yash Kale (Project Manager), Bluestock Fintech

02 Project Structure & Technologies Used

The repository is organized into clearly separated stages of the analytics workflow — raw and processed data, notebooks for each analysis phase, reusable ETL/metric scripts, SQL assets, the Power BI dashboard file, and final reporting deliverables.

Folder Structure

```
MUTUAL_FUND_PROJECT/
■■■ data/ (raw, processed, db)
■■■ notebooks/ (ingestion, cleaning, EDA, performance, advanced analytics)
■■■ scripts/ (etl_pipeline.py, compute_metrics.py, live_nav_fetch.py, recommender.py)
■■■ sql/ (schema.sql, queries.sql)
■■■ dashboard/ (Mutual_Fund_Dashboard.pbix)
■■■ reports/ (Final_Report.pdf, Presentation.pptx)
■■■ README.md, requirements.txt, .gitignore
```

Technologies Used

Category	Tools
Programming	Python, Pandas, NumPy
Database	SQLite, SQL
Analysis	Jupyter Notebook, Matplotlib, Seaborn
Visualization / BI	Power BI
Data Access	Requests API (live NAV feeds)
Version Control	Git & GitHub

03 Dataset & ETL Pipeline

Dataset

The project uses mutual fund datasets covering fund master information, NAV history, AUM details, scheme performance, and investor transactions. Processed datasets are stored under `data/processed/`.

- Fund Master
- NAV History
- AUM Details
- Scheme Performance
- Investor Transactions

ETL Pipeline

The ETL pipeline extracts raw CSV files and transforms them into an analysis-ready form before loading them into the SQLite database.

- Extract raw CSV files
- Remove duplicates
- Handle missing values
- Standardize column names
- Convert date columns
- Store cleaned data
- Load data into the SQLite database

04 Database Design

SQLite is used as the backend database, stored at `data/db/bluestock_mf.db`. It holds five core tables that support all downstream analysis and dashboarding.

Table	Purpose
`fund_master`	Master reference data for each mutual fund scheme
`nav_history`	Historical daily NAV records per scheme
`fact_aum`	Assets Under Management by fund / AMC over time
`scheme_performance`	Computed return and risk metrics per scheme
`investor_transactions`	SIP, lumpsum and redemption transaction records

05 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand data quality, distributions, and relationships across fund categories before deeper financial analysis.

Analysis Areas

- Missing Value Analysis
- Duplicate Analysis
- Distribution Analysis
- Correlation Analysis
- Fund Category Analysis
- AUM Analysis
- NAV Trends
- Performance Comparison

Visualizations Used

- Histograms
- Boxplots
- Bar Charts
- Line Charts
- Scatter Plots
- Heatmaps

06 Financial Performance Analytics

A comprehensive suite of return and risk metrics was calculated for every scheme to support fair, like-for-like comparison across the fund universe.

Metric	What it Measures
Daily Returns	Day-over-day change in NAV
CAGR	Compounded annual growth rate of the fund
Volatility	Standard deviation of returns (risk)
Sharpe Ratio	Risk-adjusted return vs. risk-free rate
Sortino Ratio	Risk-adjusted return using downside deviation
Beta	Sensitivity of fund returns to the benchmark
Alpha	Excess return over the benchmark
Tracking Error	Deviation of fund returns from its benchmark
Maximum Drawdown	Largest peak-to-trough decline
Value at Risk (VaR)	Potential loss at a given confidence level
Conditional VaR (CVaR)	Expected loss beyond the VaR threshold

07 Advanced Analytics

Beyond standard performance metrics, the project layers on advanced analytical techniques to support deeper risk assessment and personalized fund recommendations.

- **Cohort Analysis** — grouping investors/funds to study behavior and performance over time.
- **Risk Analysis** — evaluating volatility, drawdown, and downside risk across categories.
- **Fund Recommendation System** — suggesting funds aligned to an investor's risk appetite.
- **VaR & CVaR Analysis** — quantifying tail-risk exposure across the fund universe.

08 Power BI Dashboard — Industry Overview

This page provides an overall summary of the mutual fund dataset, including total funds, total AUM, average returns, and fund category distribution across Asset Management Companies.

Figure: Industry Overview dashboard — Bluestock Fintech Power BI report

Key Elements

- Total AUM: 39M crore — Total Schemes: 1,522 — Total Folios: 26.12K
- Latest SIP Inflow tracked at 31K
- AUM by Asset Management Company — led by SBI Mutual Fund, ICICI Prudential, and HDFC Mutual Fund
- Industry AUM Trend showing consistent growth from 2023 to 2025

09 Power BI Dashboard — Fund Performance

This page analyzes fund performance using CAGR, annual returns, Sharpe Ratio, and NAV trends, with interactive slicers for Fund House, Category, and Plan.

Figure: Fund Performance dashboard — Bluestock Fintech Power BI report

Key Elements

- Scheme-level table of 3-Year Return %, Sharpe Ratio, and AUM (Cr) for quick comparison
- Fund Risk vs. Return Profile bubble chart across categories (ELSS, Flexi Cap, Gilt, Large Cap, Liquid, Mid Cap, etc.)
- NAV vs. Benchmark (Nifty 50) trend line from 2022 to 2026

10 Power BI Dashboard — Investor Analytics

This page focuses on investor behavior, tracking transaction volumes, transaction types, and demographic patterns across states and age groups.

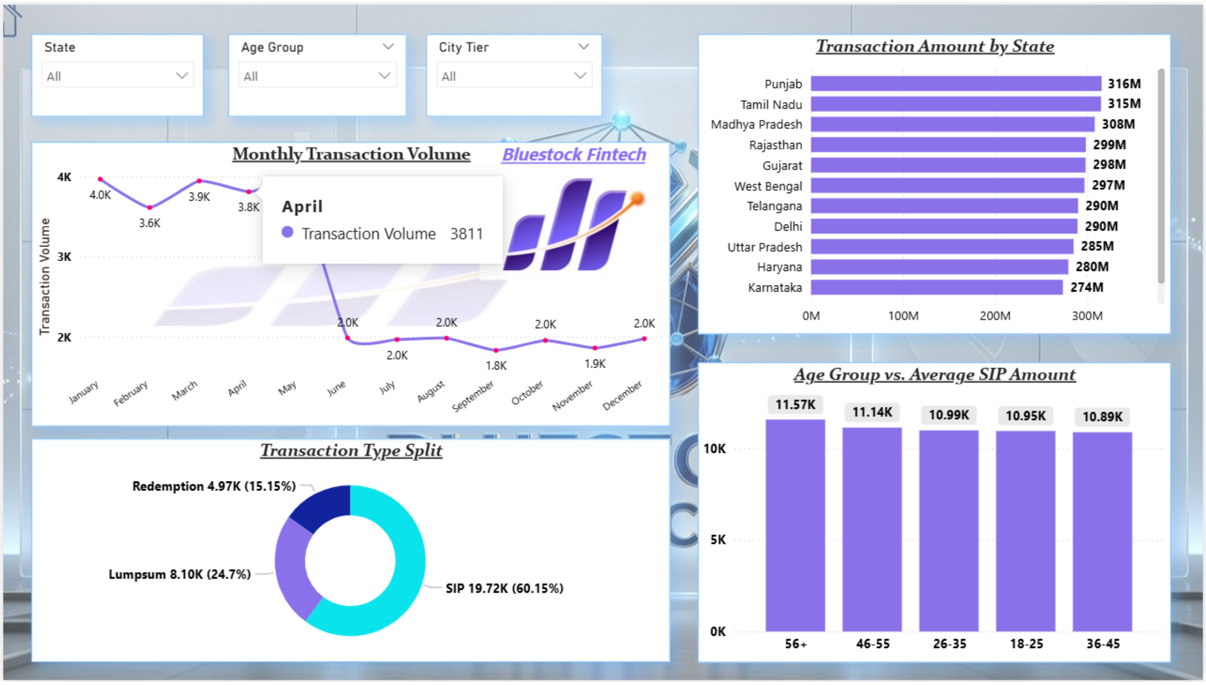

Figure: Investor Analytics dashboard — Bluestock Fintech Power BI report

Key Elements

- Monthly Transaction Volume trend, peaking near 4K in Q1
- Transaction Type Split — SIP (60.15%), Lumpsum (24.7%), Redemption (15.15%)
- Transaction Amount by State — led by Punjab, Tamil Nadu, and Madhya Pradesh
- Age Group vs. Average SIP Amount, highest among the 56+ age group

11 Power BI Dashboard — SIP & Market Trends

This page tracks Systematic Investment Plan (SIP) inflows by fund category and benchmarks them against broader market movement (Nifty 50), alongside a category-wise inflow heatmap.

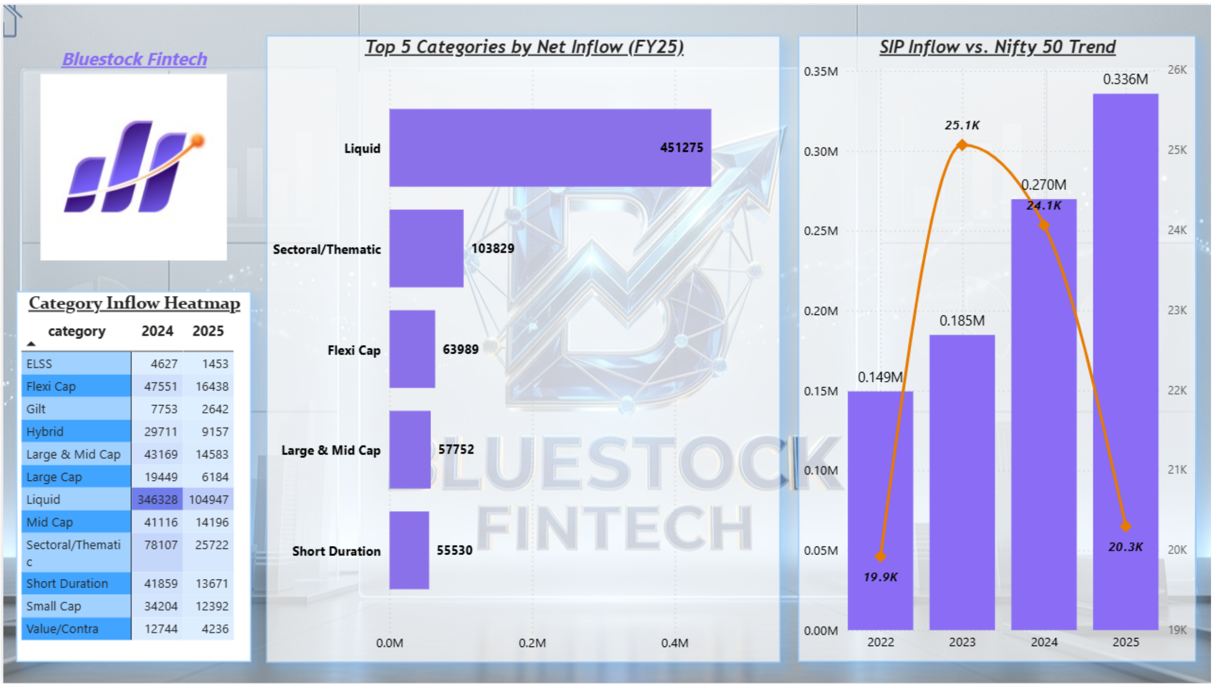

Figure: SIP & Market Trends dashboard — Bluestock Fintech Power BI report

Key Elements

- Top 5 Categories by Net Inflow (FY25) — led by Liquid funds at 451,275
- Category Inflow Heatmap comparing 2024 vs. 2025 across ELSS, Flexi Cap, Gilt, Hybrid, and more
- SIP Inflow vs. Nifty 50 Trend from 2022 to 2025, showing SIP inflows growing even through market volatility

12 SQL Queries

SQL scripts under `sql/` support table creation, data loading, and the analytical queries that feed the dashboard and reports.

- Table Creation
- Data Loading
- Fund Performance Queries
- Top Performing Funds
- Average Expense Ratio
- Highest AUM
- Risk Analysis Queries

13 Key Insights

01	Equity funds generated higher long-term returns than debt funds.
02	Lower expense ratio funds generally delivered better risk-adjusted performance.
03	Higher AUM does not always indicate superior returns.
04	Sharpe Ratio effectively identifies consistent performers across categories.
05	Maximum Drawdown highlights downside risk across funds during volatile periods.
06	SIP inflows remain resilient and continue growing even during market corrections.
07	Liquid funds dominate net inflows, reflecting investor preference for low-risk, short-term instruments.

14 How to Run the Project

Clone Repository

```
https://github.com/Shaikvali-skills/mutual_fund_project
```

Install Dependencies

```
pip install -r requirements.txt
```

Run ETL Pipeline

```
python scripts/etl_pipeline.py
```

Compute Performance Metrics

```
python scripts/compute_metrics.py
```

Fetch Latest NAV

```
python scripts/live_nav_fetch.py
```

Project Outputs

- Cleaned CSV Files
- SQLite Database
- SQL Queries
- EDA Notebooks
- Performance Analytics
- Advanced Analytics
- Power BI Dashboard
- Final Report & Presentation

15 Future Improvements & Conclusion

Future Improvements

- Live NAV updates using the MFAPI service.
- A Streamlit web application for interactive, self-serve exploration.
- Portfolio optimization using the Markowitz Model.
- Monte Carlo simulation for forward-looking risk scenarios.
- Automated email reporting for stakeholders.

Conclusion

This capstone project delivers an end-to-end mutual fund analytics solution — from raw data ingestion through to an interactive, four-page Power BI dashboard covering industry overview, fund performance, investor analytics, and SIP/market trends. The combination of rigorous financial metrics (CAGR, Sharpe, Sortino, VaR, CVaR) with visual analytics equips Bluestock Fintech stakeholders with a clear, data-driven view of the mutual fund landscape to support investor recommendations and business decisions.

Author & Submission

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Submission Date	July 10, 2026

This project was developed for educational and learning purposes as part of the Bluestock Data Analytics Capstone Project.